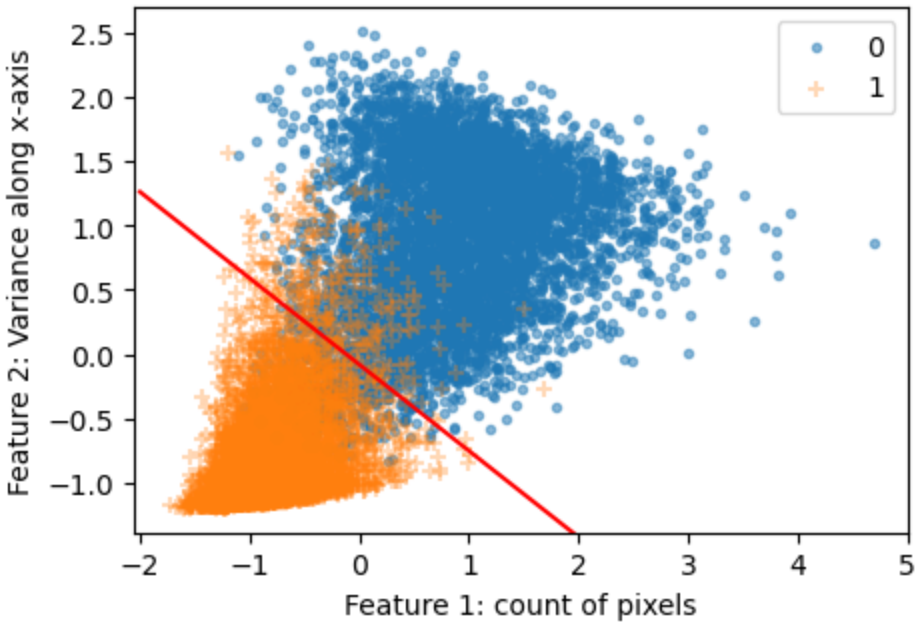
NN { Data
Model → Linear, Perceptron → Multi layer Perceptron
Loss → Least squares, Hinge loss, Cross Entropy loss
Training → Analytical → GD → SGD/Nesterov/Adam

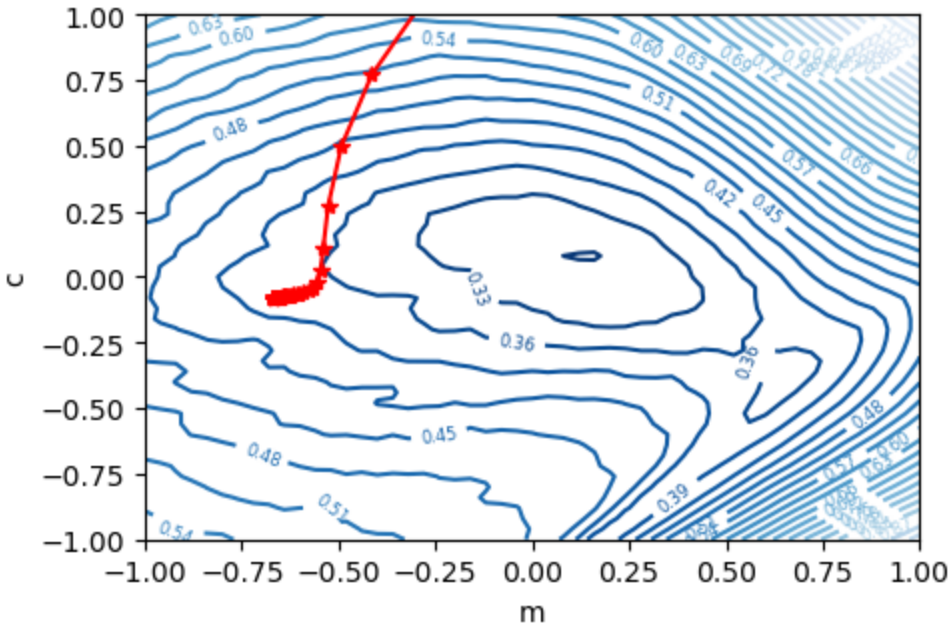
Multi layer Perceptron

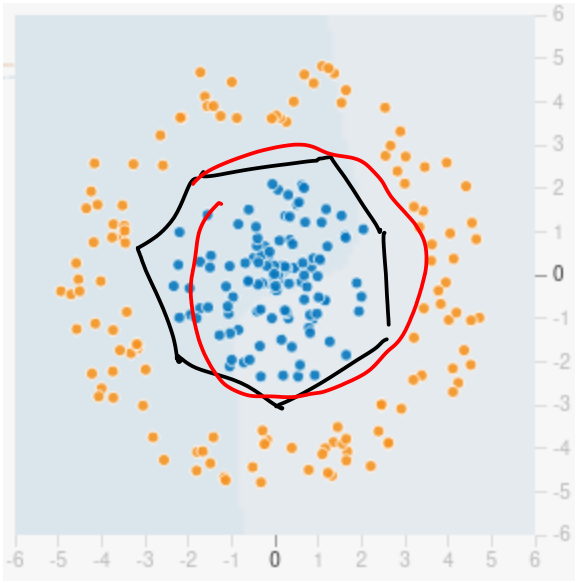
All figures are from Chapter 3 of UDLBook. https://github.com/udlbook/udlbook

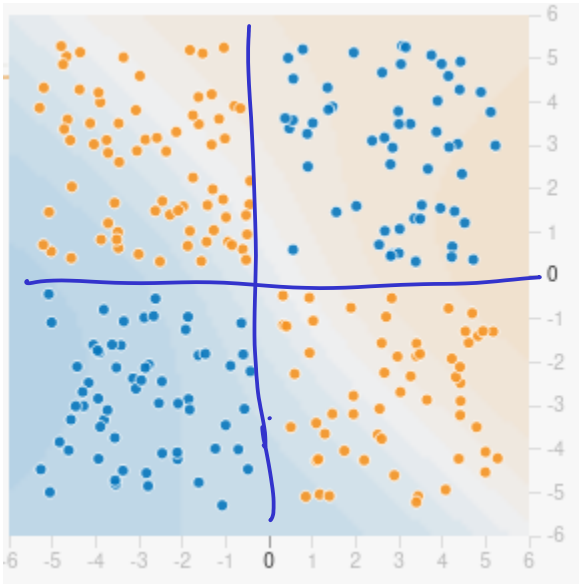
Recall the single layer perceptron

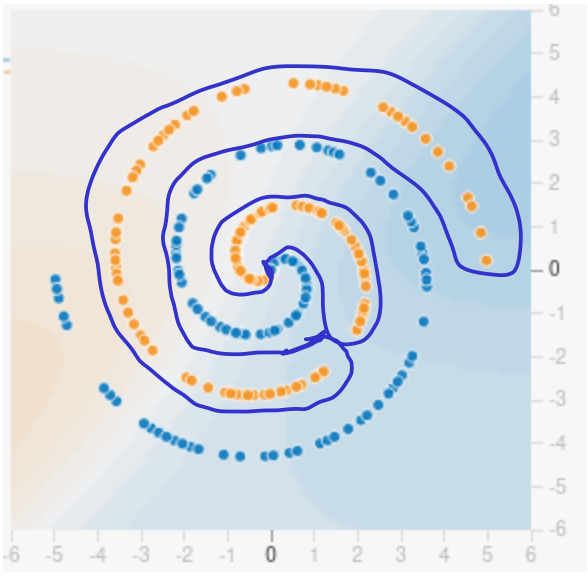


Linearly Seperable data




6 5 4 3 2 1 0 -1 -2 -3 -4 -5 -6
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6

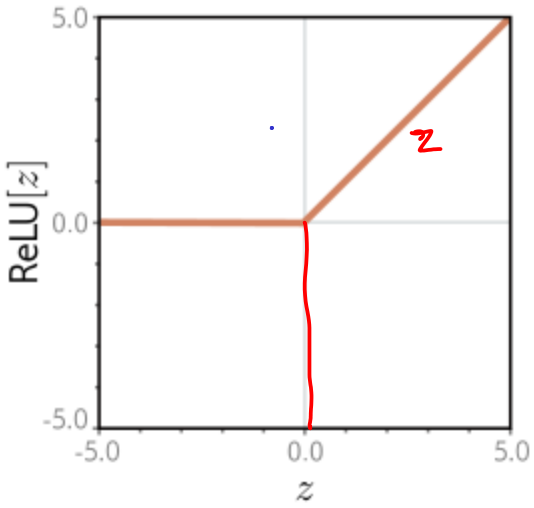

6 5 4 3 2 1 0 -1 -2 -3 -4 -5 -6
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6


6 5 4 3 2 1 0 -1 -2 -3 -4 -5 -6
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 6

$$l = f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + w_0$$ Linear

Multi Layer Perceptrons

ReLU activation function $\mathrm{ReLU}(z) = \max\{0, z\}$

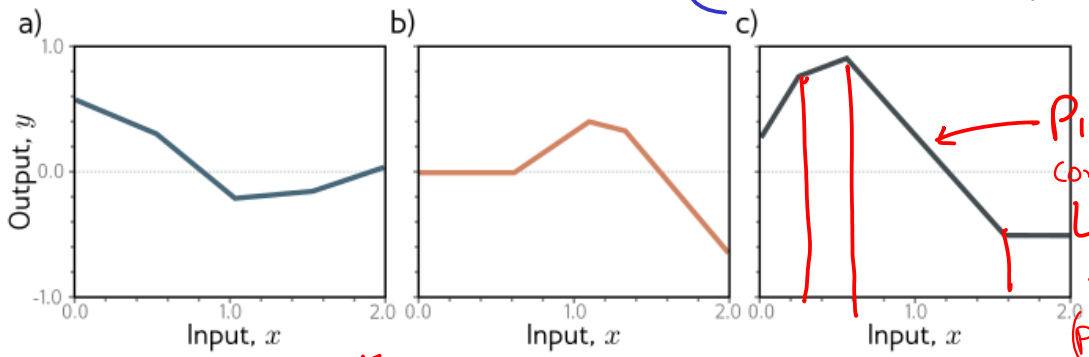


Activation function (has to be non-Linear)

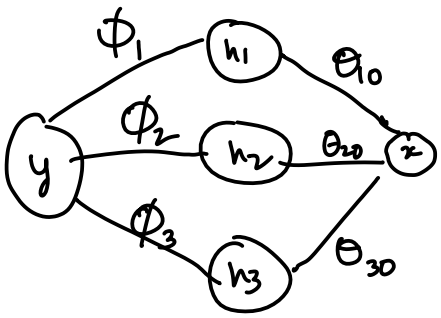
Two layer Perceptron

$$y = f(\mathbf{x}) = \mathrm{Linear}(\ \mathrm{ActivationFunction}(\ \mathrm{Linear}(x)))$$

$$\underline{w}_2^T \left(\sigma \left(\underline{w}_1 \underline{x} + \underline{b}_1 \right) \right) + b_2$$



Example



$$a(x) = \mathrm{ReLU}(x)$$

$$h_1 = a(\theta_{10} + \theta_{11} x)$$

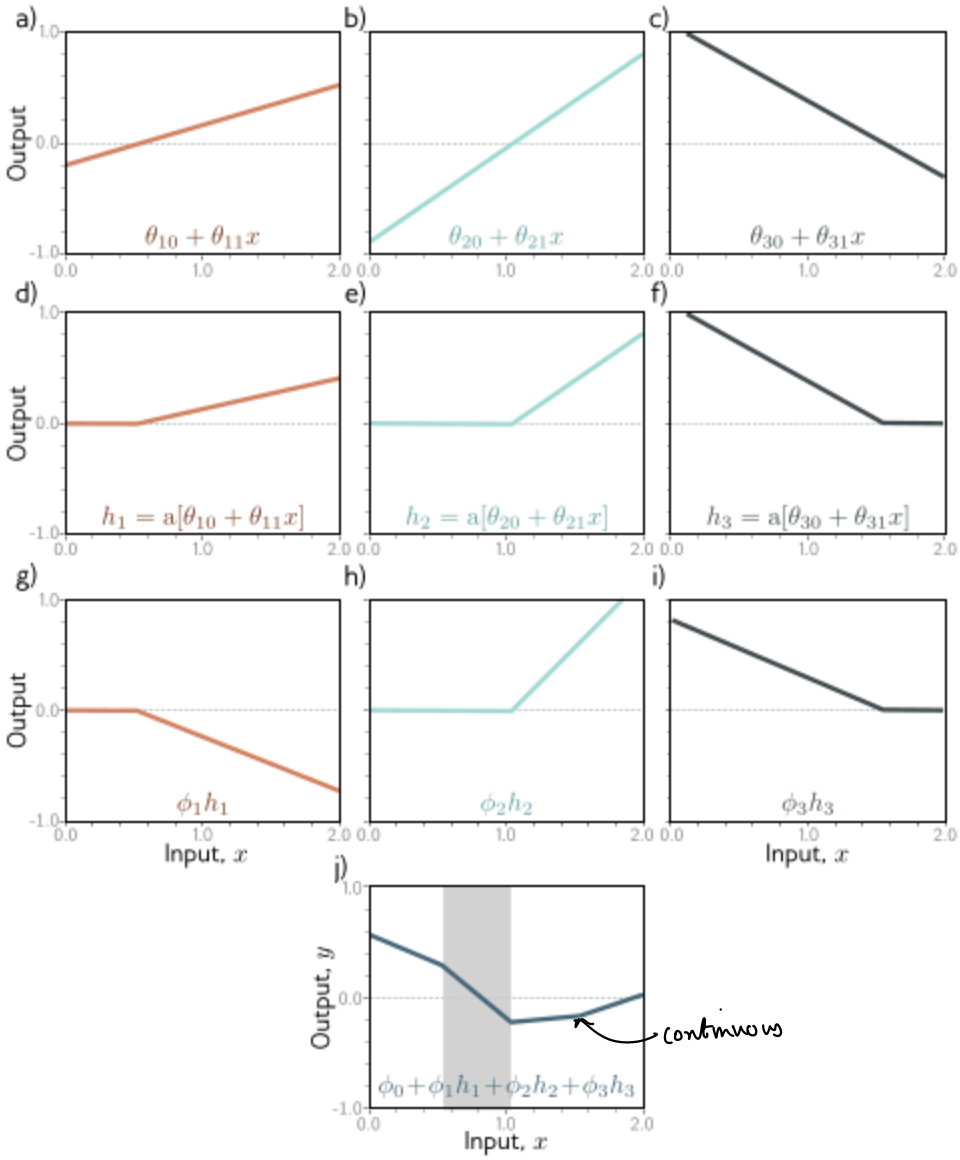
$$h_2 = a(\theta_{20} + \theta_{21} x)$$

$$y = \phi_0 + \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 \end{bmatrix} a \left(\begin{bmatrix} \theta_{11} \\ \theta_{21} \\ \theta_{31} \end{bmatrix} x + \begin{bmatrix} \theta_{10} \\ \theta_{20} \\ \theta_{30} \end{bmatrix} \right)$$

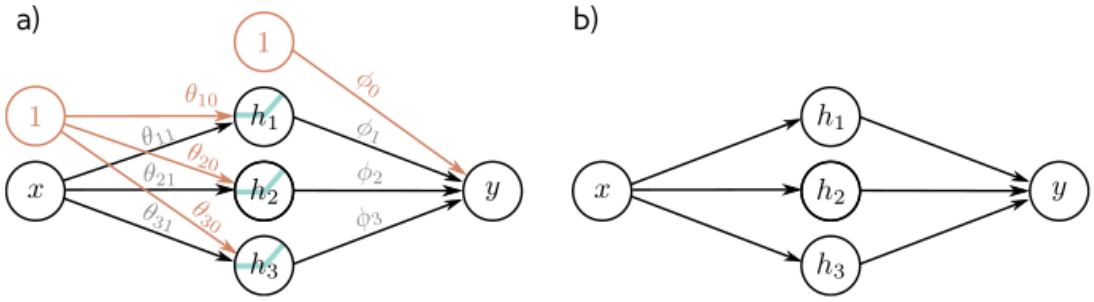
Linear

$$h_3 = a(\theta_{30} + \theta_{31}x)$$

$$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$$



Depicting Neural Networks



What if we used a linear activation function? Why is non-linear activation function important?

$$\hat{f}(x) = \overbrace{\text{Linear}(\text{Linear}(x))}^{\text{Linear}}$$

hidden units

$$\overbrace{f \circ g}^{\text{Linear}}$$

Linear Linear

$$\begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_m \end{bmatrix} = \underbrace{\begin{bmatrix} | & | & & | \\ \underline{w}_1 & \underline{w}_2 & \cdots & \underline{w}_n \\ | & | & & | \end{bmatrix}}_{W_1} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$

$$\hat{\underline{f}}(\underline{x}) = W_2(W_1\underline{x} + \underline{b}_1) + \underline{b}_2$$

$$= \underbrace{W_2 W_1}_{W'}\underline{x} + \underbrace{W_2\underline{b}_1 + \underline{b}_2}_{\underline{b}'}$$

Ranks of Matrices

true dimension

$$A = \begin{bmatrix} | & | & & | \\ \underline{a}_1 & \underline{a}_2 & \cdots & \underline{a}_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} — & \underline{a}_1^T & — \\ — & \underline{a}_2^T & — \\ & \vdots & \end{bmatrix}$$

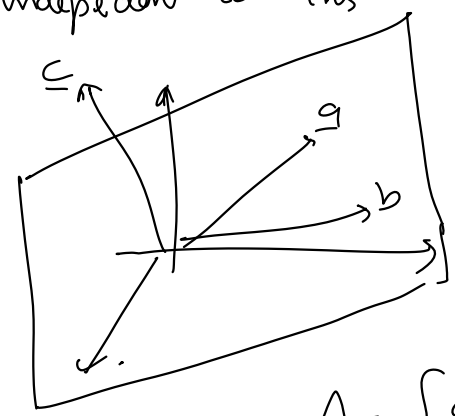
Linear independence

A vector $\underline{c}$ is LI of $\underline{a}$ and $\underline{b}$ if

$\nexists \lambda_1, \lambda_2$ such that $\underline{c} = \underbrace{\lambda_1 \underline{a} + \lambda_2 \underline{b}}_{\text{Linear combination of } \underline{a} \text{ and } \underline{b}}$

$\forall \lambda_1, \lambda_2$ the space "spanned" by $\underline{a}$ and $\underline{b}$

Number of Linearly independent columns



Less parameters ↓ more parameter ↙ ↘ parameter

$$W' = W_2 W_1$$

$$\underbrace{\text{rank}(W') = \min(\text{rank}(W_1), \text{rank}(W_2))}_{\text{less representation}}$$

$$A = \begin{bmatrix} | & | & | \\ \underline{a} & \underline{b} & \underline{c} \\ | & | & | \end{bmatrix}$$

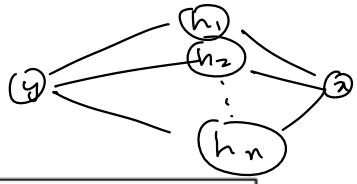
if $\underline{c}$ is LI of $\underline{a}$ and $\underline{b}$ than $\text{rank}(A) = 3$

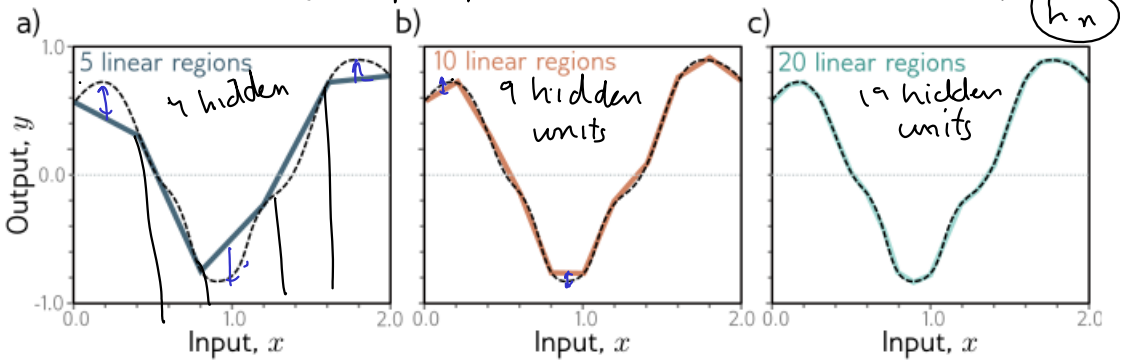
if c is not LI of $\underline{a}$ and $\underline{b}$ $\text{rank}(A) = 2$

How do you choose non-linear layer?
- Look at the data and make decision based on smoothness
- Decide based on what other people have been doing? Look for papers that have solved similar problems.
- Trial and error
- Architecture search

Universal Approximation Theorem

2-layer perceptrons.

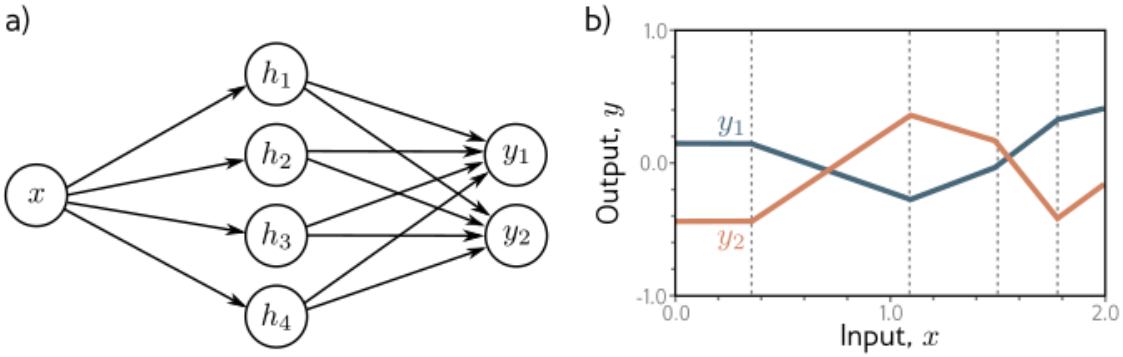


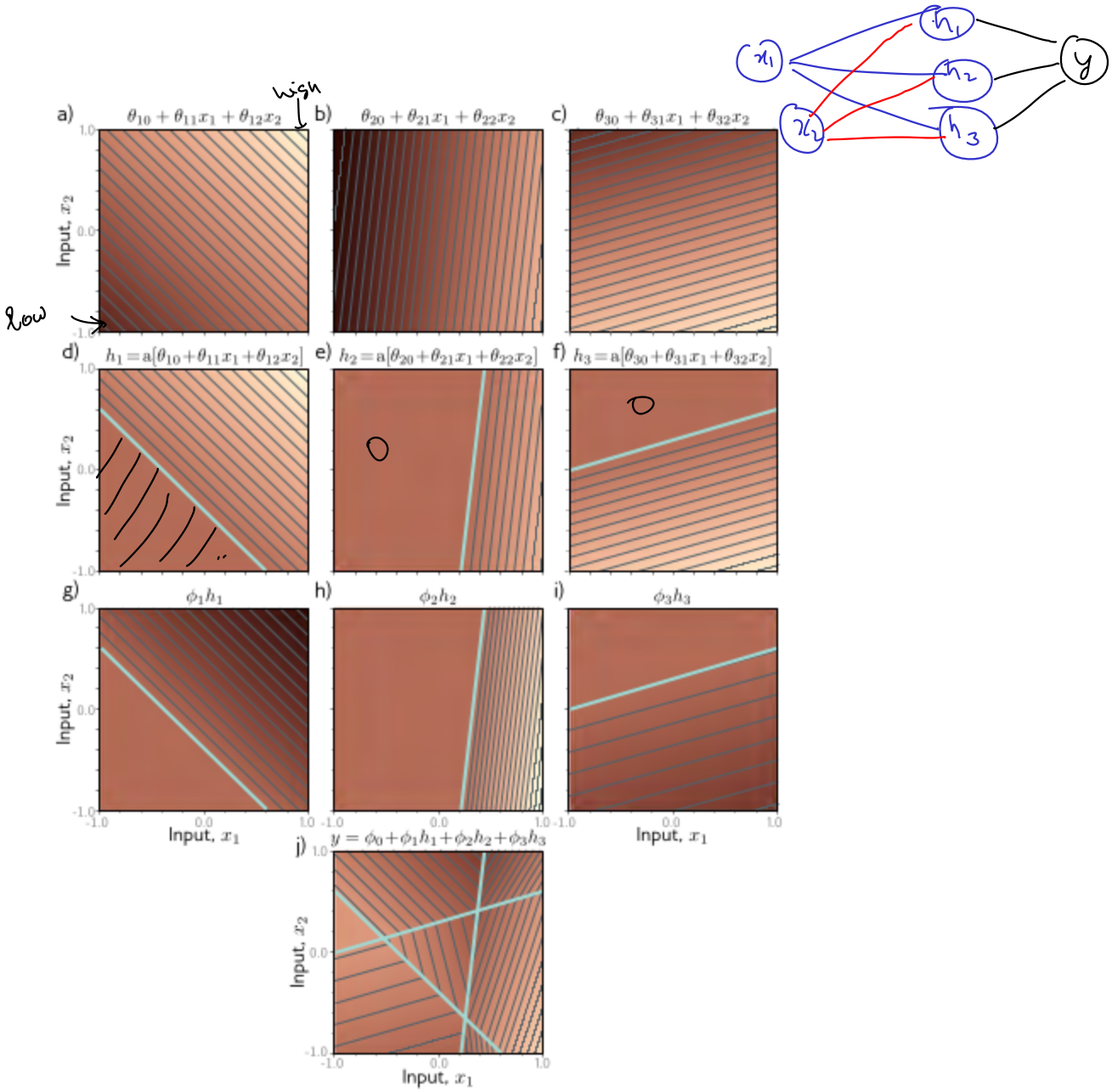


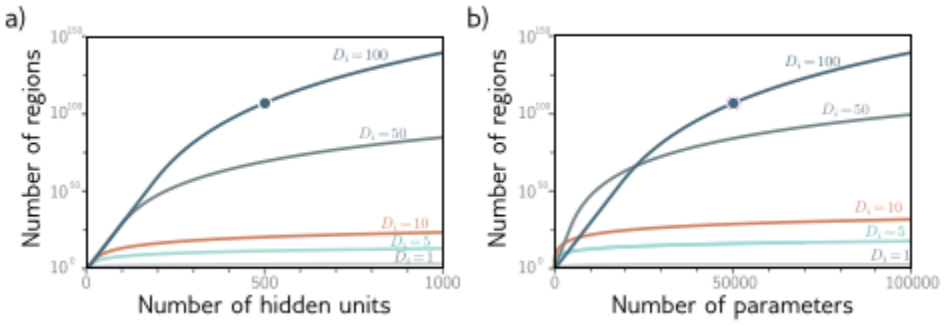
A two layer perceptron with ReLU activation function can approximate any continuous function with arbitrary amount of precision.

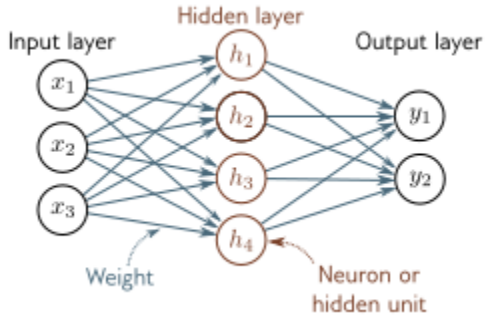
No guarantees about finding that function

Multivariate outputs

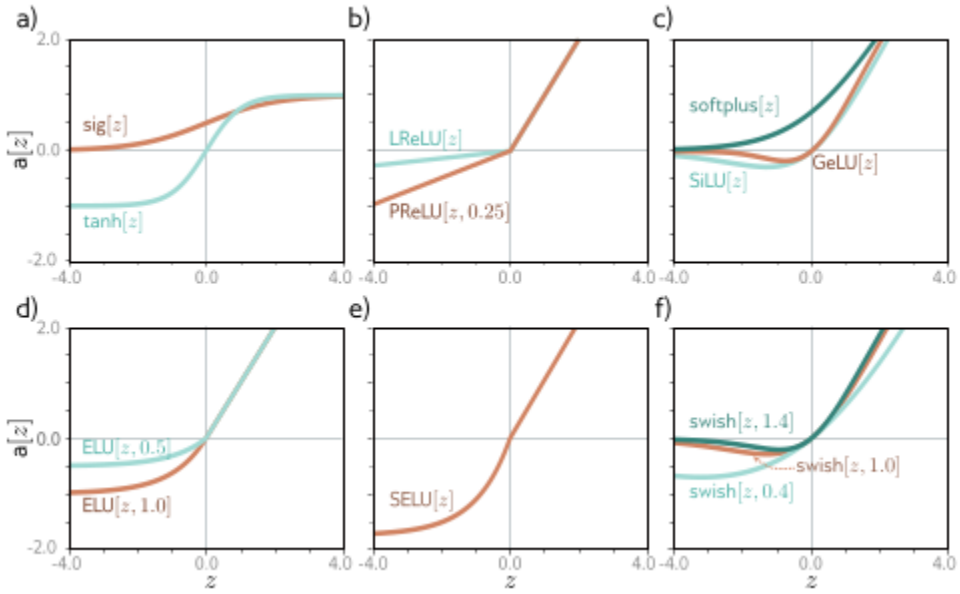


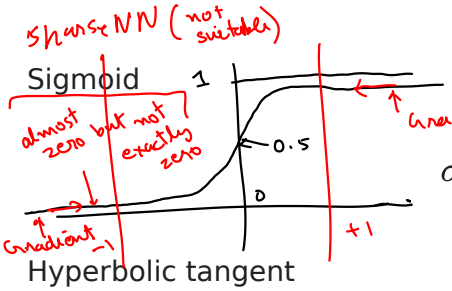

a)
$\theta_{10} + \theta_{11}x_1 + \theta_{12}x_2$
b)
$\theta_{20} + \theta_{21}x_1 + \theta_{22}x_2$
c)
$\theta_{30} + \theta_{31}x_1 + \theta_{32}x_2$
high
low
d)
$h_1 = a[\theta_{10} + \theta_{11}x_1 + \theta_{12}x_2]$
e)
$h_2 = a[\theta_{20} + \theta_{21}x_1 + \theta_{22}x_2]$
f)
$h_3 = a[\theta_{30} + \theta_{31}x_1 + \theta_{32}x_2]$
g)
$\phi_1 h_1$
h)
$\phi_2 h_2$
i)
$\phi_3 h_3$
j)
$y = \phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3$
Input, x_1
Input, x_2
x_1
x_2
h_1
h_2
h_3
y


a)
b)
Number of regions
Number of hidden units
Number of parameters
$D_i = 100$
$D_i = 50$
$D_i = 10$
$D_i = 5$
$D_i = 1$
0
500
1000
50000
100000



Activation functions



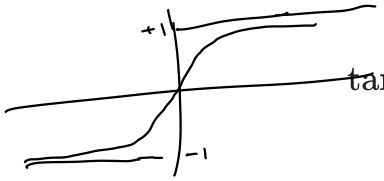


2008 - ReLU
not differentiable
differentiable

$$\sigma(z) = \frac{1}{1 + exp(-z)}$$

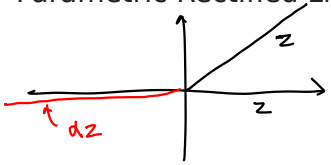
↑ expensive to compute

Hyperbolic tangent



$$\tanh(z) = \frac{exp(z) - exp(-z)}{exp(z) + exp(-z)}$$

Parametric Rectified Linear Unit



$$\text{PReLU}(z, \alpha) = \begin{cases} z & \text{if } z > 0 \\ \alpha z & \text{if } z \leq 0 \end{cases}$$

↑ very small

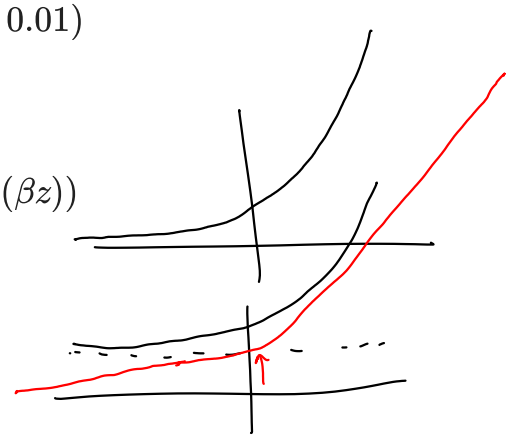
$\alpha = 0.01$
$\alpha = 0.001$
} Leaky

Leaky Rectified Linear Unit

$$\text{LReLU}(z) = \text{PReLU}(z, \alpha = 0.01)$$

Softplus

$$\text{softplus}(z) = \frac{1}{\beta}\log(1 + \exp(\beta z))$$



Gaussian error Linear Units

$$\text{GELU}(z) = z\Phi(z)$$

where $\Phi(z)$ is the error function or the cumulative distribution function of a Gaussian distribution.

Sigmoid Linear Unit

$$\text{SiLU}(z) = z\sigma(z)$$

Exponential Linear Unit

$$\text{ELU}(z, \alpha) == \begin{cases} z & \text{if } z > 0 \\ \alpha(\exp(z) - 1) & \text{if } z \leq 0 \end{cases}$$

Scaled exponential linear unit

$$\text{SELU}(z) = 1.0507 * \text{ELU}(z, 1.673)$$

Swish

$$\text{Swish}(z, \beta) = z\sigma(\beta z)$$

HardSwish

$$\text{HardSwish}(z) = \begin{cases} 0 & z < -3 \\ z(z+3)/6 & -3 \leq z \leq 3 \\ z & z > 3 \end{cases}$$